(a) Model architecture

Prediction $\hat{y}_i$

FiLM predictor
see (c)

Local condition h_i

× R

Add & Norm

Feed-forward

Add & Norm

Joint attention
see (b)

x_i

Query
e_i

Reference
K: e_j
V: $e_j + E_\gamma(y_j)$

Top-K union
using b_{rlh}

u_i

Query
(x_i, u_i)

Training bank
(x_j, u_j, y_j)

(b) Dimension-wise scores → joint attention

One block r and head h; all D coordinate branches are used

Concatenate all l,h → output projection

Context l = 1
$\Sigma_j \tilde{\alpha}_{ihj} v_{j1h}$

Context l = 2
$\Sigma_j \tilde{\alpha}_{ihj} v_{j2h}$

Context l = D
$\Sigma_j \tilde{\alpha}_{ihj} v_{jlh}$

V_1 projection

V_2 projection

V_l projection

ONE joint softmax over shared candidate IDs j
Mask → softmax → attention dropout: $\tilde{\alpha}_{ihj}$

Combine scores across all coordinates
$s = (1/D)\Sigma_l c_l + \Sigma_l g_l + \Sigma_l p_l$

l = 1
Content c_1
Bias g_1, prior p_1

l = 2
Content c_2
Bias g_2, prior p_2

l = D
Content c_l
Bias g_l, prior p_l

Q_1, K_1 projections

Q_2, K_2 projections

Q_l, K_l projections

Shared reference IDs; independent dimension projections

(c) FiLM predictor

× S layers

$h_i \rightarrow$ FiLM: γ_s, β_s

x_i

Linear

$(1+\tanh \gamma_s) \odot a_s + \beta_s$

GELU + dropout

Linear readout